S.C.O.U.T. · ECEN · ADVISOR REVIEW PACKAGE

Sensor interface review: datasheets, block diagrams and test protocol

Prepared for Dr. Shoba Krishnan, answering the three assignments from the 2026-10-07 joint advisor meeting. For every part on the buoy, this package covers:

- the datasheet lines the design relies on, as screenshots;
- what those lines mean for the circuit;
- the interface block diagram to the Feather M0;
- how it will be tested.

Full system drawing, cable schedule and parts list: S.C.O.U.T. Electronics Drawing Set SCOUT-E-100 Rev B. Values marked † are proposed and get confirmed on the bench.

DOCUMENT

Sensor Interface Review

DOC NO.	REVISION
SCOUT-E-110	A
DATE	**DUE**
2026-10-12	2026-10-14
PREPARED BY	**FOR REVIEW BY**
Isabella Rodriguez (ECEN)	Dr. Shoba Krishnan

CONTENTS

1 Assignments + sensor list · 2 System interface · 3 DS18B20 · 4 Turbidity · 5 SHT40 · 6 Adalogger (RTC + SD) · 7 Battery sense · 8 Hydrophone · 9 Power switch · 10 Test protocol

1 · What was asked, and the sensor list

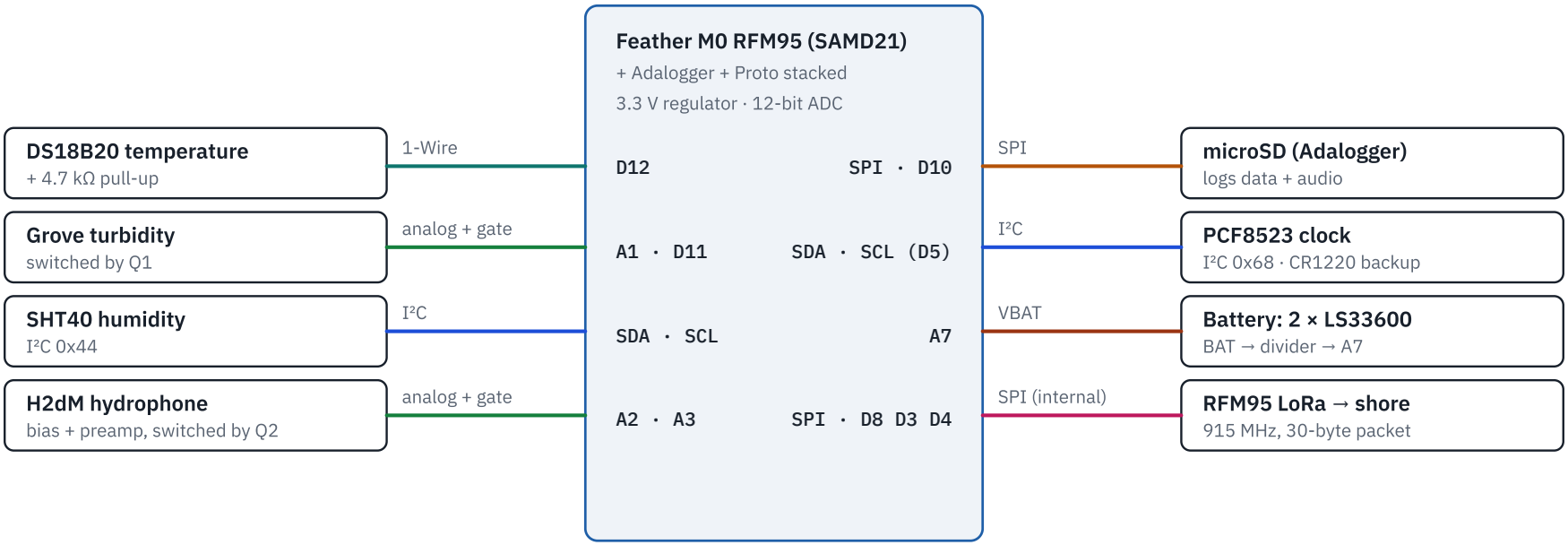
#	ASSIGNMENT (2026-10-07 ACTION ITEMS)	WHERE IT IS ANSWERED	STATUS
1	List every sensor and draw a block diagram for each (inputs, outputs, protocol, connection to the M0)	Table below; diagram on each of sheets 3–9	DONE
2	Read datasheets and app notes for every ordered sensor (pull-ups, connectors, protocol)	Screenshots and takeaways on sheets 3–9	DONE
3	Draft the sensor test protocol: one at a time, then as a packet, with averaging and timestamps	Sheet 10	DRAFT

PART	MEASURES / DOES	INTERFACE	SUPPLY	CURRENT (DATASHEET)	M0 PIN(S)	EXTRA PARTS NEEDED
DS18B20 probe (Adafruit 381)	Water temperature, ±0.5 °C	1-Wire, digital	3.0–5.5 V → 3.3 V	1–1.5 mA reading; 750 nA standby	D12	4.7 kΩ pull-up DQ→3.3 V; screw terminal
Grove turbidity (Seeed 101020752)	Water clarity	Analog voltage	3.3 V or 5 V → 3.3 V	Not published, so we measure it (Stage 2)	A1 (+ D11 switch)	P-FET switch, 100 kΩ, 100 µF + 0.1 µF
SHT40 (Adafruit 4885)	Humidity + temperature inside the housing (leak alarm)	I²C, address 0x44	3.3–5 V → 3.3 V	0.4 µA average, 80 nA idle	SDA, SCL	None: 10 kΩ pull-ups on board
PCF8523 RTC (Adalogger 2922)	Timestamps	I²C, 0x68; INT open-drain	3.3 V + CR1220 backup	Small; measured in Stage 2	SDA, SCL, (D5)	CR1220 cell; INT wire to D5 if used
microSD (Adalogger 2922)	Data + audio log	SPI, CS = D10	3.3 V	Write peaks measured in Stage 2	SCK, MOSI, MISO, D10	None (100 kΩ CS pull-up on board)
Battery sense (on the Feather)	Battery voltage	Analog, 2:1 divider	VBAT	18 µA continuous (3.6 V / 200 kΩ)	D9 = A7	None
Aquarian H2dM hydrophone	Reef sound	Analog, 3.5 mm plug-in power	≥ 2 V, 0.7 mA bias	≈ 1.1 mA with the preamp, only while recording	A2 (+ A3 switch)	1.1 kΩ bias, MCP6002 preamp, P-FET switch

PART	MEASURES / DOES	INTERFACE	SUPPLY	CURRENT (DATASHEET)	M0 PIN(S)	EXTRA PARTS NEEDED
RFM95 LoRa (on the Feather)	Sends the data packet to shore	SPI, CS D8 · IRQ D3 · RST D4	3.3 V	Measured in Stage 2 (+11 dBm)	`internal`	Antenna: spring (bench), uFL→SMA whip (deployed)

2 · System interface at a glance

Everything runs from the Feather's 3.3 V regulator; there is no 5 V rail. Sensors that draw real current sit behind a P-channel switch that the M0 turns on only while it reads them. All grounds are common.



Shared buses: I²C carries SHT40 (0x44) and PCF8523 (0x68), so the addresses are unique. SPI carries microSD (CS D10) and LoRa (CS D8), and only one CS is LOW at a time.

3 · DS18B20 waterproof temperature probe

ADVISOR POINT · "The DS18B20 is 1-Wire and needs a pull-up; it does not just plug into the M0." Confirmed below from the datasheet.

FEATURES

- Unique 1-Wire® Interface Requires Only One Port Pin for Communication
- Each Device has a Unique 64-Bit Serial Code Stored in an On-Board ROM
- Multidrop Capability Simplifies Distributed Temperature-Sensing Applications
- Requires No External Components
- Can Be Powered from Data Line; Power Supply Range is 3.0V to 5.5V
- Measures Temperatures from -55°C to +125°C (-67°F to +257°F)
- ±0.5°C Accuracy from -10°C to +85°C
- Thermometer Resolution is User Selectable from 9 to 12 Bits
- Converts Temperature to 12-Bit Digital Word in 750ms (Max)

Maxim DS18B20 datasheet, Features. One data pin (1-Wire). Supply 3.0–5.5 V, so 3.3 V is valid. ±0.5 °C from −10 to +85 °C. 750 ms max at 12 bits.

Figure 5. Powering the DS18B20 with an External Supply

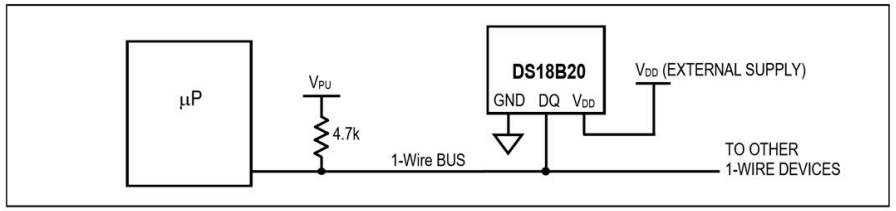


Figure 5, Powering with an external supply. This is the wiring we use: VDD to 3.3 V (not parasite power), and a 4.7 kΩ pull-up from DQ to V_{PU}.

DC ELECTRICAL CHARACTERISTICS (-55°C to +125°C; V_{DD}=3.0V to 5.5V)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
Supply Voltage	V_{DD}	Local Power	+3.0		+5.5	V	1
Pullup Supply Voltage	V_{PU}	Parasite Power	+3.0		+5.5	V	1,2
		Local Power	+3.0		V_{DD}		
Thermometer Error	t_{ERR}	-10°C to +85°C			±0.5	°C	3
		-55°C to +125°C			±2		
Input Logic-Low	V_{IL}		-0.3		+0.8	V	1,4,5
Input Logic-High	V_{IH}	Local Power	+2.2		The lower of 5.5 or V_{DD} + 0.3	V	1, 6
		Parasite Power	+3.0				
Sink Current	I_L	$V_{I/O}$ = 0.4V	4.0			mA	1
Standby Current	I_{DDS}			750	1000	nA	7,8
Active Current	I_{DD}	V_{DD} = 5V		1	1.5	mA	9
DQ Input Current	I_{DQ}			5		µA	10
Drift				±0.2		°C	11

DC Electrical Characteristics. V_{IH} min 2.2 V (local power), so the M0's 3.3 V logic is fine. Active current 1–1.5 mA, standby 750–1000 nA.

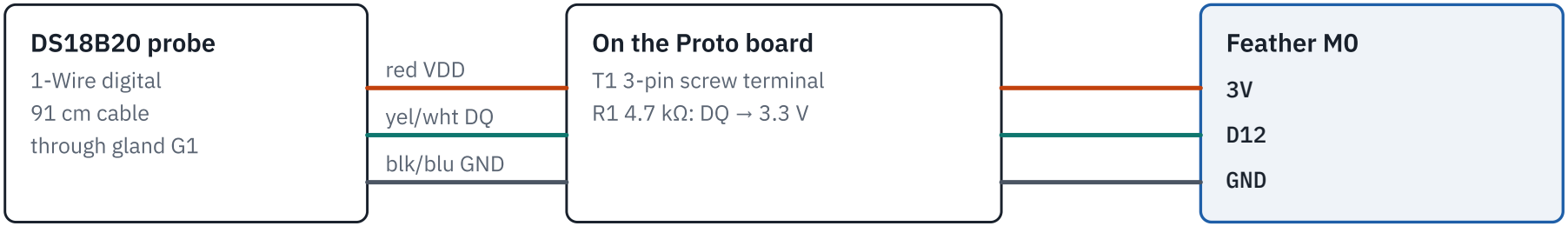
Table 2. Thermometer Resolution Configuration

R1	R0	RESOLUTION (BITS)	MAX CONVERSION TIME	
0	0	9	93.75ms	(t_{CONV}/8)
0	1	10	187.5ms	(t_{CONV}/4)
1	0	11	375ms	(t_{CONV}/2)
1	1	12	750ms	(t_{CONV})

Table 2, Resolution. 12-bit takes up to 750 ms. Firmware must wait for it, or drop to 11-bit (375 ms) to save awake time.

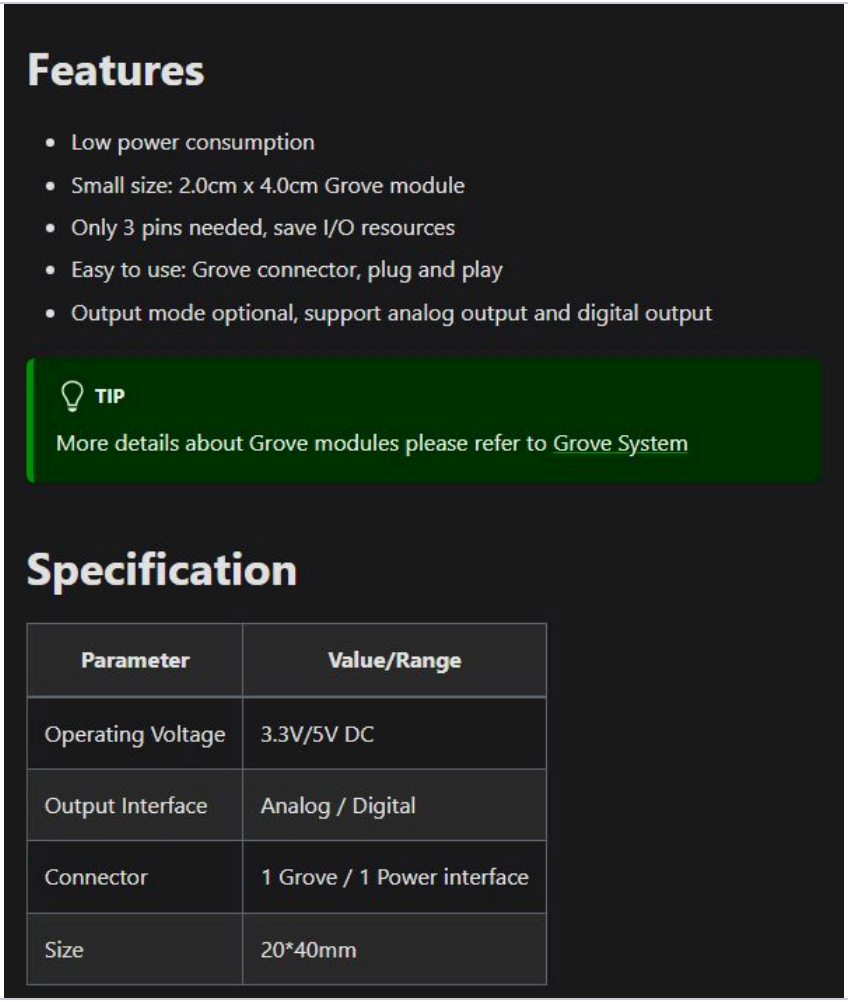
What it means for our circuit

- Powered from 3.3 V on its own VDD wire, not parasite power, so the 1-Wire timing stays simple.
- 4.7 kΩ pull-up from DQ to 3.3 V. It comes in the Adafruit bag and is soldered on the Proto board.
- Standby is under 1 µA, so it does not need a power switch.
- Wires: red VDD, yellow/white DQ, black/blue GND (depends on batch). They go into a 3-pin screw terminal.



Firmware: OneWire + DallasTemperature, start conversion, wait 750 ms, read with CRC check.

4 · Grove turbidity sensor (Seeed 101020752)



Features

- Low power consumption
- Small size: 2.0cm x 4.0cm Grove module
- Only 3 pins needed, save I/O resources
- Easy to use: Grove connector, plug and play
- Output mode optional, support analog output and digital output

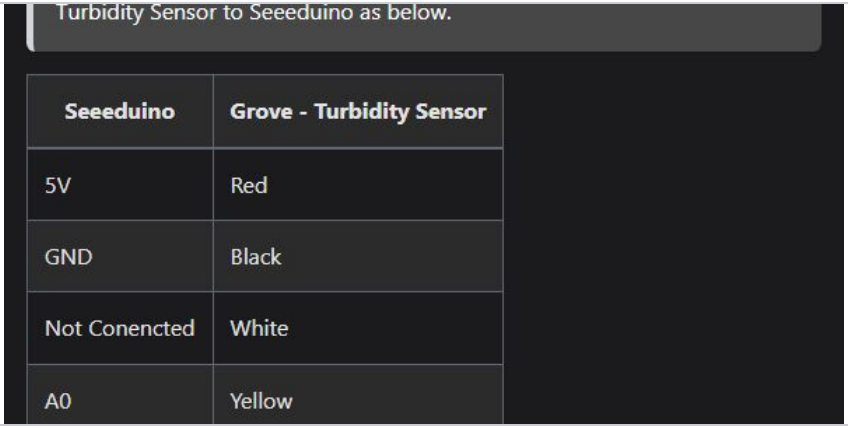
TIP

More details about Grove modules please refer to Grove System

Specification

Parameter	Value/Range
Operating Voltage	3.3V/5V DC
Output Interface	Analog / Digital
Connector	1 Grove / 1 Power interface
Size	20*40mm

Seeed wiki, Specification. Operating voltage 3.3 V/5 V DC, so it runs on our 3.3 V rail with no boost and no divider. Analog or digital output.



Turbidity Sensor to Seeeduino as below.

Seeeduino	Grove - Turbidity Sensor
5V	Red
GND	Black
Not Conencted	White
A0	Yellow

Seeed wiki, Hardware connection. Red VCC, black GND, white not connected, yellow signal to an analog pin. Their example shows 5 V; we use 3.3 V, which the spec table allows.

Grove - Turbidity Sensor Meter for Arduino V1.0



The Grove turbidity sensor can measure the turbidity of the water (the number of suspended particles).

The optical sensor of this module can measure the density of turbid water and the concentration of extraneous matter using the refraction of wavelength between photo transistor and diode. By using an optical transistor and optical diodes, an optical sensor measures the amount of light coming from the source of the light to the light receiver, in order to calculate turbidity of water.

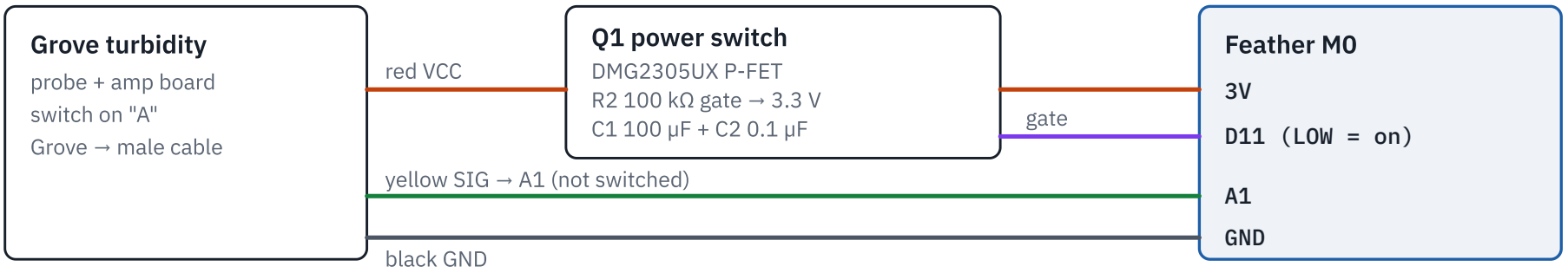
The output mode can be selected by adjusting the switch on the board. Supports analog and digital output. The sensitivity can be adjusted by the on-board knob.

Get One Now

Seeed wiki, Description. Optical: an LED and a phototransistor measure light through the water. The output mode is set by the on-board switch (we use analog), and the knob sets sensitivity.

What it means for our circuit

- Runs on 3.3 V, so the output stays inside the ADC range and goes straight to A1 with no divider. That replaces the old 5 V SEN0189 and its level shifting.
- The LED draws continuous current and the wiki gives no current figure. So it goes behind P-FET switch Q1 on D11 (sheet 9) and is powered only while reading. Current is measured in Stage 2 †.
- Output falls as turbidity rises. Calibrate a clear-water baseline (the "zero") before deployment.
- Firmware waits for the reading to settle after power-on (500 ms in config; to be measured †).



5 · SHT40 humidity and temperature (Adafruit 4885)

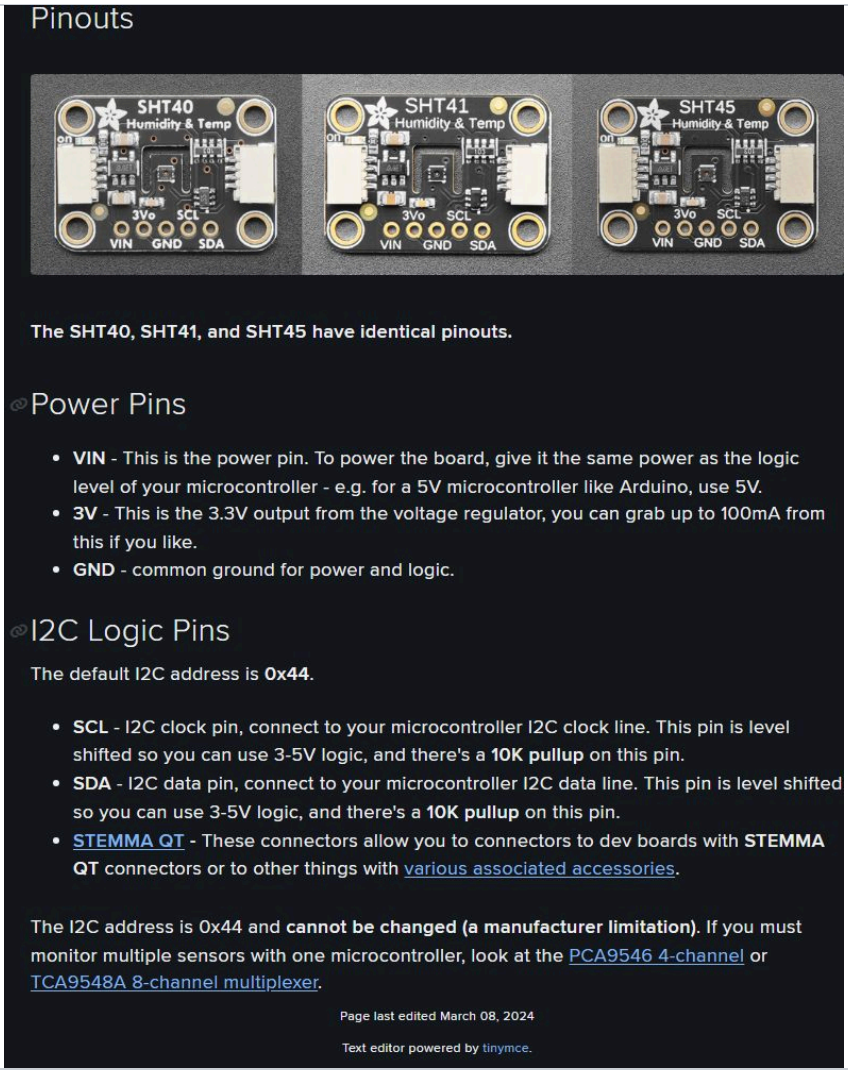
Revision History:

- **As of October 2022** – we've updated this PCB with Adafruit Pinguin to make a lovely and legible silkscreen - you may get the new PCB or the older version with vector fonts - both are identical other than the fancy silkscreen. Additionally, this sensor breakout may come with black or tan STEMMA QT connectors. They work the same!

SHT40 Specifications & Features

- Relative humidity accuracy: up to ±1.8 %RH
- Temperature accuracy: up to ±0.2 °C
- Breakout supply voltage: 3.3 to 5V
- Average bare sensor current: 0.4 µA (at meas. rate 1 Hz)
- Idle bare sensor current: 80 nA
- I2C fast mode plus, CRC checksum
- Operating range: 0...100 %RH, -40...125 °C
- Fully functional in condensing environment
- Variable power heater
- NIST traceability for sensor
- JEDEC JESD47 qualification for sensor
- Mature technology from global market leader Sensirion
- I2C address 0x44

Adafruit product page, SHT40 specifications. ±1.8 %RH, ±0.2 °C. 0.4 µA average, 80 nA idle, so it needs no power switch. I²C with CRC checksum, address 0x44.

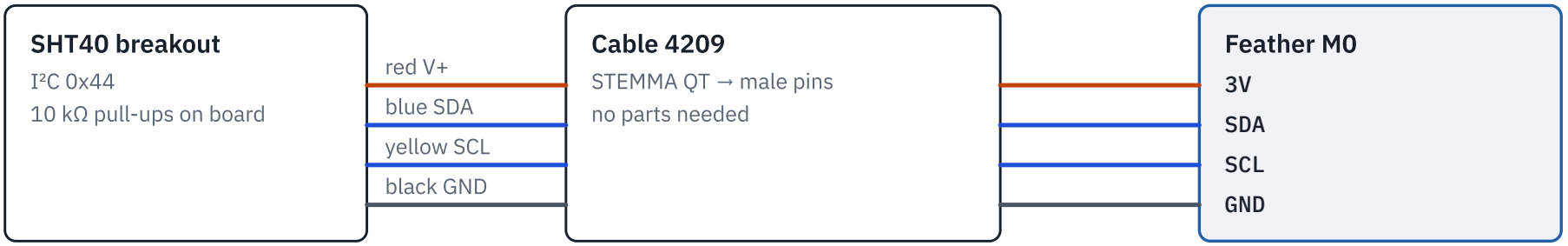


Adafruit guide, Pinouts. SDA and SCL each have a 10 kΩ pull-up on the board. The address is 0x44 and cannot be changed. STEMMA QT connector.

What it means for our circuit

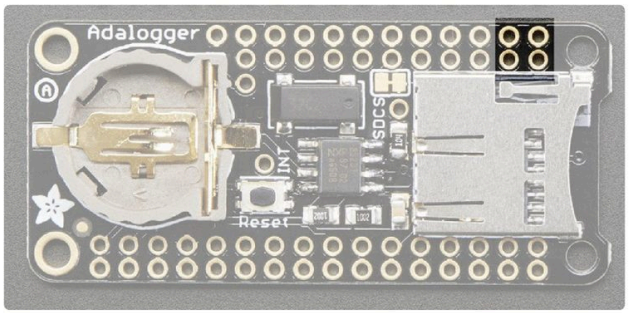
- No extra parts. The STEMMA QT → male cable (4209) goes to 3V, GND, SDA and SCL.
- It shares I²C with the PCF8523 (0x68), and the addresses don't clash. Both boards have pull-ups, so the bus has two 10 kΩ in parallel (5 kΩ). That's fine at 3.3 V.

- Job: leak and condensation alarm inside the sealed housing. A rising humidity trend flags a leak before water reaches the boards.



6 · Adalogger FeatherWing: PCF8523 clock and microSD (Adafruit 2922)

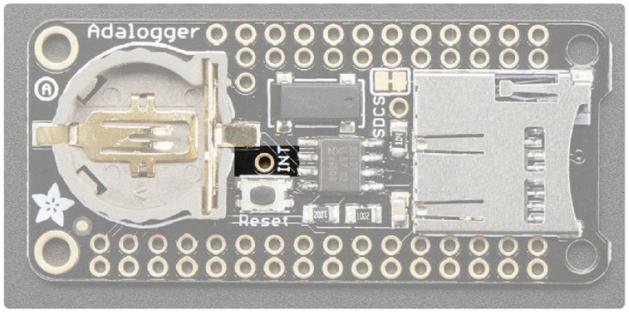
RTC & I2C Pins



In the top right, **SDA** (rightmost) and **SCL** (to the left of SDA) are used to talk to the RTC chip.

- **SCL** - I2C clock pin, connect to your microcontrollers I2C clock line. This pin has a 10K pullup resistor to 3.3V
- **SDA** - I2C data pin, connect to your microcontrollers I2C data line. This pin has a 10K pullup resistor to 3.3V

These pins are in the same location on every Feather

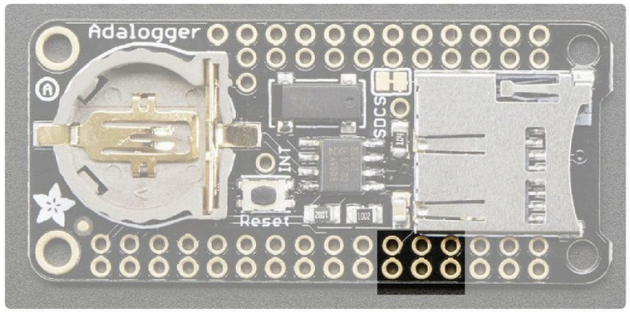


There's also a breakout for **INT** which is the output pin from the RTC. It can be used as an interrupt output or it could also be used to generate a square wave.

Note that this pin is **open drain** - you must enable the internal pullup on whatever digital pin it is connected to!

Adafruit Adalogger guide, RTC & I2C pins. SDA/SCL with 10 kΩ pull-ups to 3.3 V. INT is open-drain, so the internal pull-up must be enabled on the pin it connects to (D5 in firmware).

SD & SPI Pins



Starting from the left you've got

- **SPI Clock** (SCK) - output from feather to wing
- **SPI M**icrocontroller **O**ut **S**ensor **I**n (MOSI) - output from feather to wing
- **SPI M**icrocontroller **I**n **S**ensor **O**ut (MISO) - input from wing to feather

These pins are in the same location on every Feather. They are used for communicating with the SD card. When the SD card is not inserted, these pins are completely free. MISO is tri-stated whenever the **SD CS** pin is pulled high

The **SDCS** pin is the chip select line.

- On ESP8266, the SD CS pin is on GPIO **15**
- On ESP32 it's GPIO **33**
- On WICED it's GPIO **PB5**
- On the nRF52832 it's GPIO **11**
- On Atmel M0, M4, 328p or 32u4 it's on GPIO **10**
- On Teensy 3.x it's on GPIO **10**

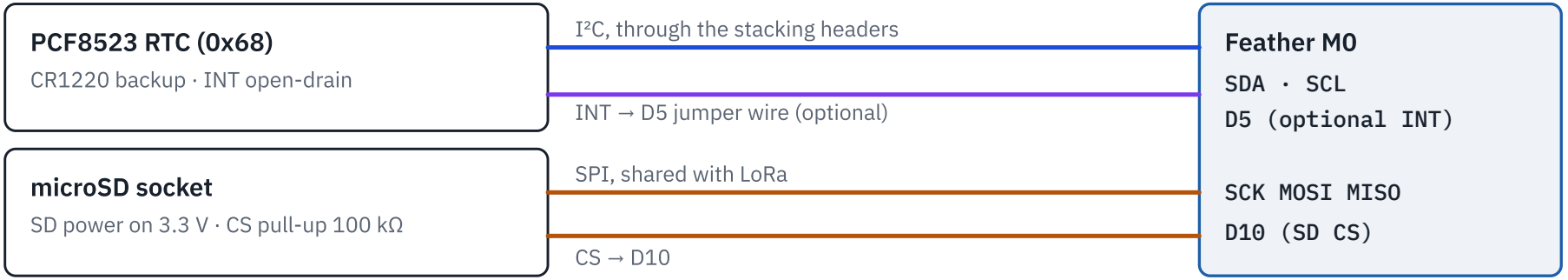
Some users report ESP32 variants may use GPIO10 instead. For the FeatherS3, users report GPIO3.

You can cut the trace to the default pin and change this to any pin by soldering a wire

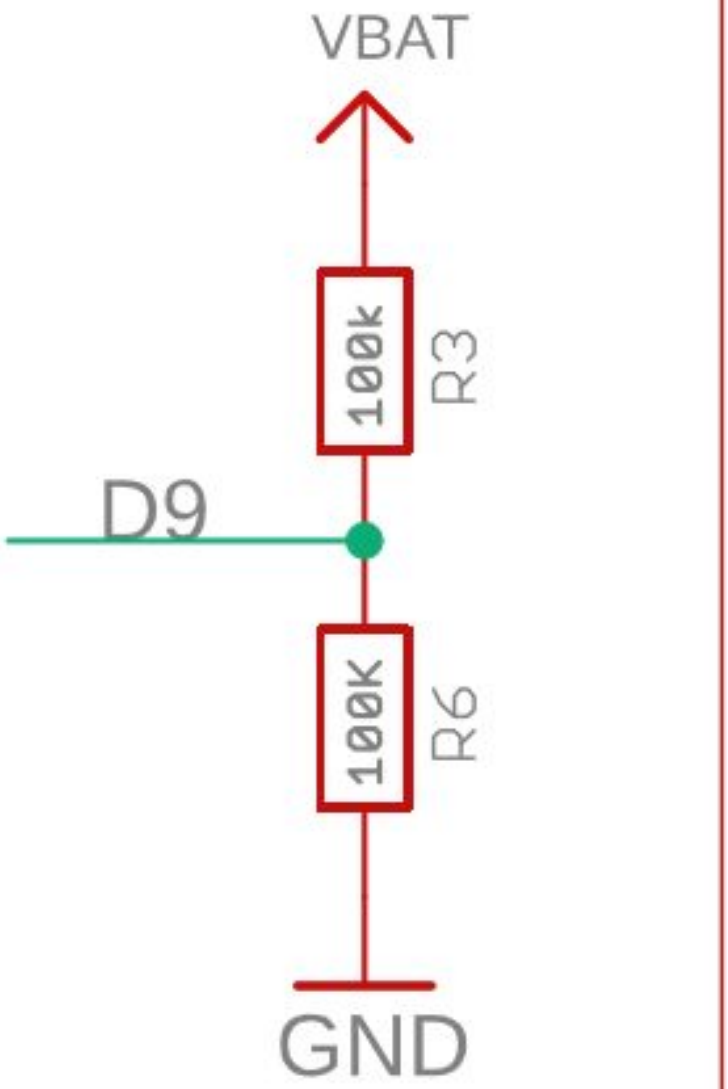
Adafruit Adalogger guide, SD & SPI pins. Shares the hardware SPI pins. "On Atmel M0 ... it's on GPIO 10", so SD CS = D10. MISO tri-states when CS is high, which is why it can share SPI with the radio.

What it means for our circuit

- No wiring needed: it plugs onto the M0 with stacking headers. Add the CR1220 cell so the clock keeps time without power.
- Every record is timestamped from the PCF8523. That's the "add a timestamp" step from the advisor feedback.
- SPI is shared with the LoRa radio (CS D8). Firmware keeps D8 HIGH while using the SD card, and D10 HIGH while using the radio.
- Optional: one jumper wire from INT to D5 lets the clock wake the M0. The current plan wakes on the SAMD21's internal RTC, so this is a backup.



7 · Battery voltage sense (on the Feather M0)



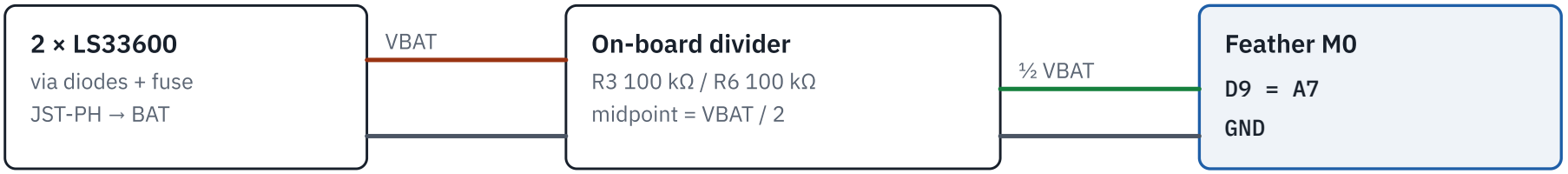
Adafruit Feather M0 schematic. R3 and R6, both 100 kΩ, from VBAT to GND, with the midpoint on D9.



Same schematic, MCU pin table. D9 is PA07 = AIN7, which is why firmware reads it as A7.

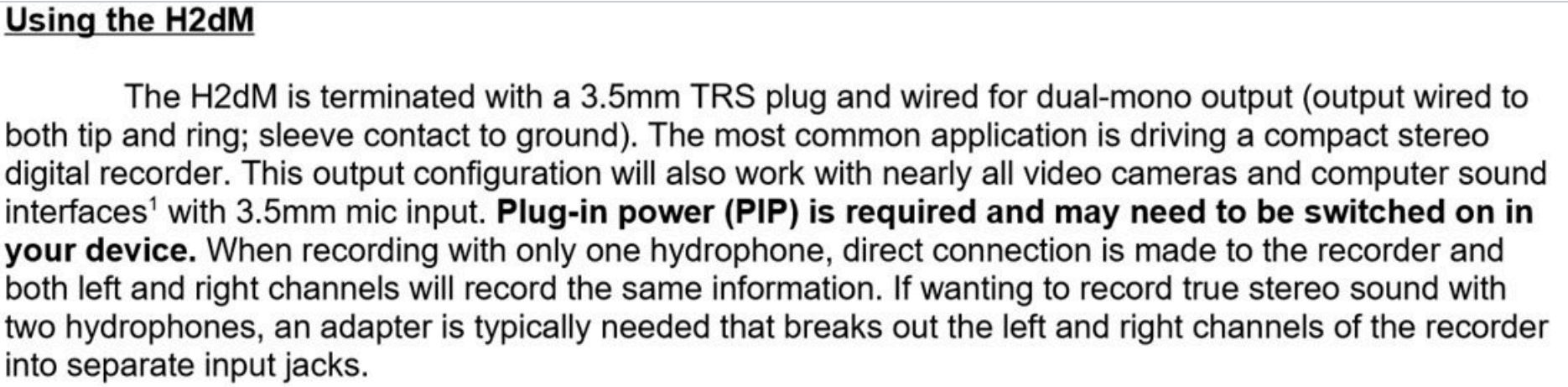
- V_{bat} = ADC × 2 × 3.3 V / 4096 (12-bit). At 3.6 V the pin sits at 1.8 V, inside the ADC range.
- The divider always draws 3.6 V / 200 kΩ = 18 µA. That's counted in the power budget.

- Used for the firmware's power tiers: normal ≥ 3.2 V, conserve 3.1–3.2 V, critical < 3.1 V (provisional).



8 · Aquarian H2dM hydrophone (planned purchase, 6 m cable)

OPEN FROM 2026-10-09 · The meeting note suggested regulating down from a larger battery. Our plan keeps the hydrophone on the 3.3 V rail, because the manual below only asks for 2 V of plug-in power.



Using the H2dM

The H2dM is terminated with a 3.5mm TRS plug and wired for dual-mono output (output wired to both tip and ring; sleeve contact to ground). The most common application is driving a compact stereo digital recorder. This output configuration will also work with nearly all video cameras and computer sound interfaces[1] with 3.5mm mic input. **Plug-in power (PIP) is required and may need to be switched on in your device.** When recording with only one hydrophone, direct connection is made to the recorder and both left and right channels will record the same information. If wanting to record true stereo sound with two hydrophones, an adapter is typically needed that breaks out the left and right channels of the recorder into separate input jacks.

Aquarian H2dM User's Guide p. 1. 3.5 mm TRS, dual-mono: the same output on tip and ring, sleeve to ground. "Plug-in power (PIP) is required", so we must supply the bias ourselves.

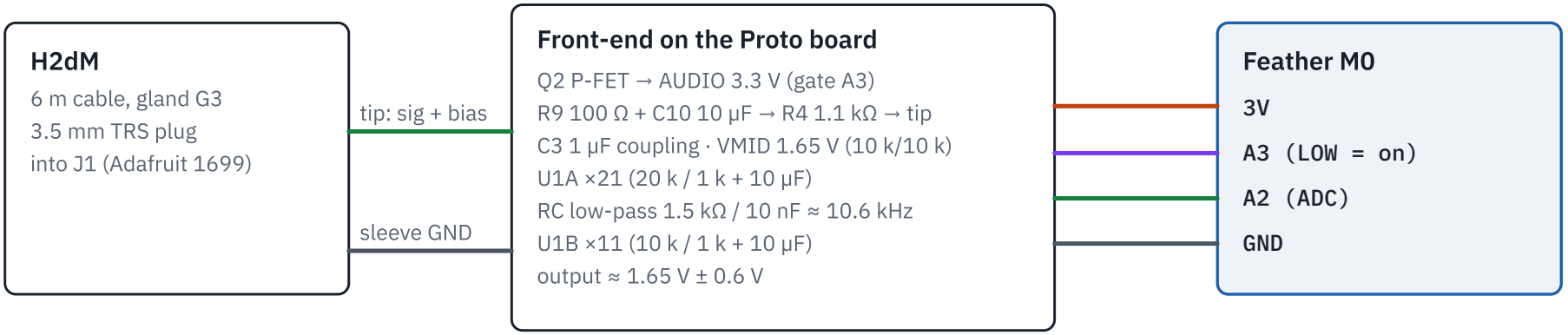
Note that there is no standard for plug-in power. Performance can vary significantly. The following specifications are based on typical response when using the H2dM with a compliance voltage of 2.5V and bias resistor of 1.1KΩ (two channels at 2.2KΩ each—2.2mA—typical of many compact digital recorders, including those that we sell).

Sensitivity:	-172dB re: 1V/µPa	(+/- 4dB 20Hz-4KHz)
Useful range:	<10 Hz to >100KHz	(not measured above 100KHz, approximate sensitivity @100KHz = -220dB re: 1V/µPa)
Polar Response:	Omnidirectional (horizontal)	
Operating depth:	<80 meters	
Output impedance:	1.1 KΩ	(typical)
Power:	0.7 mA	(typical)
Minimum PIP req.:	2V, 0.7mA	(compliance voltage, short-circuit current) (the hydrophone will work with lower current, but with very unpredictable gain and greatly-reduced dynamic range)
Physical:		(cable and output plug excluded)
Dimensions:	25mm x 46mm	
Mass:	105 grams	
Specific Gravity:	5.3	

Same guide p. 2, Specifications. –172 dB re 1 V/µPa. Output impedance 1.1 kΩ. 0.7 mA. **Minimum PIP 2 V, 0.7 mA.** Specs are rated at a 2.5 V compliance voltage through a 1.1 kΩ bias resistor, which is the circuit we copy.

What it means for our circuit

- Bias: 3.3 V through R9 100 Ω (noise filter, with C10 10 µF) and R4 1.1 kΩ. At 0.7 mA the tip sits at 3.3 − 0.7 mA × 1.2 kΩ ≈ 2.46 V, above the 2 V minimum. No higher-voltage battery needed.
- Signal: 120 dB re 1 µPa (a typical reef) × −172 dB gives $10^{(120-172)/20}$ ≈ 2.5 mV. Two non-inverting MCP6002 stages (×21, ×11 ≈ 231, 47 dB) take it to ≈ 0.6 V around a 1.65 V mid-rail †.
- Anti-alias low-pass at ≈ 10.6 kHz (1.5 kΩ, 10 nF), for 22.05 kHz sampling on A2. One-minute clips go to SD; only summary numbers go over LoRa.
- Powered only while recording (Q2 on A3). About 1.1 mA while on: hydrophone 0.7 mA + op-amp + mid-rail divider.



Ring carries the same signal (dual-mono) and is left unconnected. Gain and filter values marked † are confirmed with a tap test on the bench.

9 · Sensor power switch: DMG2305UX P-channel MOSFET

P-CHANNEL ENHANCEMENT MODE MOSFET

Product Summary

BV_{DSS}	$R_{DS(on)}$ Max	Package	I_D Max T_A = +25°C
-20V	52mΩ @V_{GS} = -4.5V	SOT23	-5.0A
	100mΩ @V_{GS} = -2.5V		-3.6A

Features

- Low On-Resistance
- Low Input Capacitance
- Fast Switching Speed
- **Totally Lead-Free & Fully RoHS Compliant (Notes 1 & 2)**

Diodes Inc. DMG2305UX datasheet (DS36196) p. 1. P-channel enhancement MOSFET, −20 V, SOT-23, $R_{DS(on)}$ ≤ 100 mΩ at V_{GS} = −2.5 V.

Electrical Characteristics (@ T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Min	Typ	Max	Unit	Test Condition
OFF CHARACTERISTICS (Note 7)						
Drain-Source Breakdown Voltage	BV_{DSS}	-20	—	—	V	V_{GS} = 0V, I_D = -250μA
Zero Gate Voltage Drain Current T_J = +25°C	I_{DSS}	—	—	-1.0	μA	V_{DS} = -20V, V_{GS} = 0V
Gate-Source Leakage	I_{GSS}	—	—	±100	nA	V_{GS} = ±8V, V_{DS} = 0V
ON CHARACTERISTICS (Note 7)						
Gate Threshold Voltage	$V_{GS(th)}$	-0.5	—	-0.9	V	V_{DS} = V_{GS}, I_D = -250μA
Static Drain-Source On-Resistance	$R_{DS(on)}$	—	40	52	mΩ	V_{GS} = -4.5V, I_D = -4.2A
			52	100		V_{GS} = -2.5V, I_D = -3.4A
			68	200		V_{GS} = -1.8V, I_D = -2A

Same datasheet p. 2. $V_{GS(th)}$ −0.5 to −0.9 V, so a 3.3 V logic pin turns it fully on. I_{DSS} ≤ 1 μA, so leakage while off is negligible.

- High-side switch: the source goes to 3.3 V and the drain feeds the sensor. Sensor ground stays connected, so the analog readings stay referenced to ground.
- A 100 kΩ gate pull-up holds it OFF during sleep, boot and reset. Gate pin LOW = sensor ON. Firmware polarity is being updated to match (Linear SCO-186).
- At about 20 mA of sensor current the drop is under 2 mV (20 mA × 0.1 Ω), which is negligible.
- Two are used: Q1 on D11 (turbidity) and Q2 on A3 (hydrophone front-end).

10 · Sensor test protocol (draft)

> **ADVISOR METHOD** · Test one sensor at a time, then all together. Send each sensor's data to a receiver, average several readings, add a timestamp, then send all sensors as one packet. Watch for conflicts when combined.

For every sensor, in order

1. **Wire alone** on the breadboard, per its block diagram. Check supply and pull-up voltages with the meter before connecting the M0.
2. **Read 20 samples** over USB serial, each with an RTC timestamp. Record the mean, standard deviation, min and max.
3. **Check accuracy** against a reference (table below). Log the result in the build/test plan.
4. **Measure current** in each state (off, warming up, reading) with the meter in series, for the power budget.
5. **Send it over the radio** to the shore station and confirm the received value matches the logged one.

SENSOR	SETUP	REFERENCE / STIMULUS	PASS CRITERION	RECORDS
DS18B20	3.3 V, 4.7 kΩ, D12	Ice bath (0 °C) and room-temperature water vs a reference thermometer	Within ±0.5 °C of the reference (datasheet); 0 CRC errors in 100 reads	Offset at each point, conversion time
Turbidity	Through Q1, A1	Clear water, then kaolin-clouded water	Voltage drops as cloudiness rises; 20-sample standard deviation small and repeatable †	Clear-water baseline ("zero"), settle time after power-on, current
SHT40	I²C 0x44	Room hygrometer; breath on the sensor	Within the room reference ±1.8 %RH (datasheet); RH rises then recovers after breath	Baseline RH inside the sealed housing (leak threshold)
PCF8523 + SD	Adalogger stacked	Set from the laptop clock; 24 h run	Drift measured over 24 h; 1,000 lines written and read back with no errors	Drift per day, SD write current peak
Battery sense	A7	Bench supply at 3.0 / 3.3 / 3.6 V vs the meter	Computed V_{bat} within 2 % of the meter †	Calibration factor
Hydrophone	Q2 on A3, preamp → A2	Bucket of water, tap the side; quiet baseline	A2 idles at ≈ 1.65 V; taps visible with no clipping at 0 or 3.3 V; 1-minute clip plays back	Noise floor, gain check, clip file size

All together

1. **Combine one at a time:** add sensors one by one and rerun each check. This catches I²C/SPI clashes and supply dips; scope the 3.3 V rail when Q1 and Q2 switch on.
2. **One cycle:** wake → power sensors → average → timestamp → write SD → build one 30-byte packet (David's schema) → transmit → power off → sleep.
3. **24-hour run** on the 30-minute cycle: compare the shore log with the SD log line by line. Count missing packets and check that timestamps are in order.
4. **Calibration record before deployment:** zero the turbidity in clear water, record the temperature offset, set the RTC to UTC.

Sources: Maxim DS18B20 datasheet · Seeed Grove Turbidity wiki · Adafruit SHT40 product page and guide · Adafruit Adalogger FeatherWing guide · Adafruit Feather M0 RFM9x schematic · Aquarian H2dM User's Guide · Diodes Inc. DMG2305UX datasheet DS36196. Local copies in the team DATASHEETS folder.